EDMS 23 – 301 – 1

TECHNICAL SPECIFICATION

FOR

Electrical Energy Storage System (EESS)

Issue: April. - 2024

1. SCOPE

This specification covers the minimum requirements for mobile emergency battery energy storage vehicle ⁄ stationary battery energy storage system. The design, engineering, manufacture, testing at the manufacturer's factory & field site, and delivery at the destination shall be supported.

2. APPLICABLE STANDARDS

Unless otherwise specified in this specification shall comply with the latest edition of international standards and shall be designed, manufactured, and tested in accordance to the latest applicable international standards as follows:

Table (1)

Standard No.	Description
IEC62933-1	Electrical energy storage (EES) systems - Vocabulary
IEC62933-2-1	Electrical energy storage (EES) systems – Unit parameters and testing method – General specification
IEC62933-2-2	Electrical energy storage (EES) systems – Unit parameters and testing method – Application and performance
IEC62133-2	Secondary cells and batteries containing alkaline or other non-acid electrolytes – Safety requirements for portable sealed secondary cells, and for batteries made from them, for use in portable applications – Part 2: Lithium systems
IEC 62477-1	2022 Safety requirements for power electronic converter systems and equipment - Part 1: General
IEC 60050-692-2017	International electro technical vocabulary - Part 692: Generation, transmission and distribution of electrical energy - Dependability and quality of service of electric power system
IEC62619	Secondary cells and batteries containing alkaline or other non -acid electrolytes – Safety requirements for secondary lithium cells and batteries, for use in industrial applications

IEC 62620	2023 Secondary cells and batteries containing alkaline or other non-acid electrolytes - Secondary lithium cells and batteries for use in industrial applications
NFPA 68-2018	Standard on Explosion Protection by Deflagration Venting
NFPA 69-2019	Standard on Explosion Prevention System
NFPA 855-2023	Standard for the Installation of Stationary Energy Storage Systems
UL	Safety Standard

3. ENVIRONMENTAL CONDITIONS

The performance of the system shall be guaranteed for the following environmental conditions, any differences in the guaranteed performance shall be clearly set out in the offer:

Table (2)

Minimum ambient temperature	-5°C
Maximum ambient temperature	45°C (50 °c as option).
Maximum relative humidity	95%.
Maximum altitude	1000

4. THE APPLICATION SCENARIOS OF BESS (OPTION AS … EDC):

- Scenario 1: Restore power of power outage area

Considering that the power outage area is uncertain or not fixed, mobile energy storage vehicles are often used in this scenario. Battery energy storage system is used to restore power to blackout area. In this scenario, mobile emergency battery energy storage vehicle / stationary battery energy storage system is equipped with the following functions:

- an essential function of off-grid black start, an optional function of on-grid and off-grid switching modes (if configured, switching time should be less than or equal to 20ms.)
- noise level should be less than or equal to 65dB, an optional function of coordinating operation with diesel generators, an optional function of communicating with third-party SCADA systems

- Scenario 2:Provide non-power outage services for high-reliability loads

Considering that this kind of load is “Category 1 load on the grid" in the power grid, the load is relatively clear, and diesel generators are generally equipped as standard as emergency power sources. Mobile or stationary Energy storage is often supplement load connected to the power distribution to achieve non-stop power outages. Mobile energy storage is compatible with diesel generator to guarantee high load needs.

Mobile emergency battery energy storage vehicle / stationary battery energy storage system is used to guarantee power supply for extremely critical facilities. Energy storage system is equipped with the following functions:

- an essential function of off-grid black start, an essential function of on-grid and off-grid switching should be less than or equal to 20ms
- the noise level should be less than or equal to 60dB, an essential function of coordinating operation with diesel generators, an essential function of communicating with third-party SCADA systems

- Scenario 3: Application of micro-grids and considers scenarios in conjunction with photovoltaics

This scenario can be considered to use a fixed type (no connection with the large power grid, or completely off-grid), or a mobile type (a microgrid connected to the grid, the energy storage can be withdrawn when connected to the grid, and the energy storage can be connected when off-grid) .

Mobile emergency battery energy storage vehicle / stationary battery energy storage system is used to coordinate operation with a microgrid. Battery energy storage system is equipped with the following functions:

- an essential function of off-grid black start,
- an optional function of on-grid and off-grid switching modes (if configured, switching time should be less than or equal to 20ms.),
- the noise level should be less than or equal to 65dB,
- an optional function of coordinating operation with diesel generators,
- an optional function of communicating with third-party SCADA systems,

5. DESIGN CRETERIA:

5.1 BATTERY ENERGY STORAGE SYSTEM(BESS):

Generally BESS includes a battery system, power conversion system or hybrid inverter, battery management system, environmental controls, energy management system and safety equipment such as fire suppression, sensors and alarms.

The mobile BESS is connected to the power grid for operation. When the external power supply fails, the switching device switches within 5 milliseconds maximum.

All system shall be supported with Earthing terminal for safety purpose, DC side, AC side, PCS, Batteries Rack, and so on.

5.1.1 BATTERY SYSTEM:

The battery system comprises a fixed number of lithium cells wired in series and parallel within a frame to create a module. The modules are then stacked and combined to form a battery rack. Battery racks can be connected in series or parallel to reach the required voltage and current of the battery energy storage system.

The State Of Health (SOH) shall be 100 % @ Beginning of Life (BOL).

The State of Health (SOH) shall be 80 % @ End of Life (EOL)

In Case of using electrical energy storage system as stationary a general safety consideration according IEC 62619 shall be followed for batteries:

- The battery shall comply with the general safety considerations outlined in IEC 62619
- Insulation and wiring: Ensure that wiring and insulation can handle maximum anticipated voltage, current, temperature, altitude, and humidity, with adequate clearances and creepage distances between conductors.
- Venting: The inclusion of a pressure relief function in the casing of cells, modules, battery packs, and battery systems is imperative to prevent rupture or explosion, while ensuring that the choice of encapsulation materials and methods does not impede pressure relief or lead to overheating during normal operation.
- Temperature/voltage/current management: Battery designs must prevent abnormal temperature increases, while battery systems should adhere to voltage, current, and temperature limits set by the cell manufacturer.
- Terminal contacts of the battery pack and/or battery system shall be designed to ensure secure and reliable connections with external devices.
- Assembly of cells, modules, or battery packs into battery systems should adhere to specific rules to mitigate risks effectively.
- Operating region of lithium cells and battery systems for safe use should be specified by manufacturers.

The battery must be type tested in accordance with the following table, sourced from IEC 62619.

Category	Test	Cell	Battery System
Product Safety Test (safety of cell and battery system)	External short-circuit test	√	-
	Impact test	√	-
	Drop test	√	√
	Thermal abuse test	√	-
	Overcharge test	√	-
	Forced discharge test	√	-
	Consideration of internal short-circuit (select one from the two options) - Internal short-circuit test - Propagation test	√	√
Functional safety test (safety of battery system)	Overcharge control of voltage	-	√
	Overcharge control of current	-	√
	Overheating control	-	√

In Case of using mobile electrical energy storage system, a general safety consideration according IEC 62133-2 shall be followed for batteries:

- The battery shall comply with the general safety considerations outlined in IEC 62133-2
- Insulation and Wiring:

  - The insulation resistance between the positive terminal and externally exposed metal surfaces, excluding electrical contact surfaces, shall be not less than 5 MΩ at 500 V DC when measured 60 seconds after applying the voltage.

  - Internal wiring and insulation shall be sufficient to withstand the maximum anticipated current, voltage, and temperature requirements.
- Venting: The battery shall be designed to allow for proper venting to prevent pressure build-up in case of abnormal conditions.

- Temperature, Voltage, and Current Management: The battery shall incorporate appropriate mechanisms for managing temperature, voltage, and current to ensure safe operation under normal and foreseeable conditions.
- Terminal Contacts shall be designed to ensure secure and reliable connections with external devices.
- Assembly of Cells into Batteries shall accomplish integrity and safety of the battery and adequate mechanical protection shall be provided for cells and components of the battery.

The battery must be type tested in accordance with the following table, according to IEC 62133-2

Test	Cell	Battery
Continuous Charge	5	-
Case Stress	-	3
External Short-Circuit	5 per temperature	-
External Short-Circuit	-	-5
Free Fall	3	3
Thermal Abuse	5 per temperature	-
Crush	5 per temperature	-
Overcharge	-	-5
Forced Discharge	5	-
Mechanical - Vibration - Mechanical Shock	 - -	3 3
Forced Internal Short	5 per temperature	-

5.1.2 POWER CONVERSION SYSTEM (PCS) OR HYBRID INVERTER:

This device can convert power bi-directionally. This means DC power from the battery can be converted to AC power for use with grid or electrical loads, and AC power can be converted to DC power to charge the battery. This effectively gives the EESS its ability to both charge and discharge.

The power conversion system consists of three main components: an AC circuit breaker, a converter, and a DC circuit breaker.

- AC Circuit Breaker: This component will disconnect the power unit from the utility source when necessary.
- Converter: In the event of a utility source interruption, the three-phase converter will supply power to the islanded load until utility service is restored or until the energy stored in the battery pack is depleted.
- DC Circuit Breaker: This breaker will isolate the battery pack, allowing routine maintenance to be conducted on the PCS.

The PCS (Power Conversion System) for grid-connected and islanded (off-grid) operation shall encompass the following functions:

- Efficient Bidirectional Power Transfer: The PCS must efficiently convert AC power from the battery arrays' DC power and transfer it Bi-directionally to or from the distribution line, ensuring that harmonics do not exceed the limits specified by national grid connection code.
- Black Start Capability: The PCS should be capable of initiating operation independently in case of a blackout.
- The PCS shall have the ability to synchronize its voltage and frequency with any other remote AC bus or generator during online startup.
- The PCS should demonstrate stability against typical fluctuations in grid voltage and frequency.

PCS Electrical Protection involves protecting the power conversion system from various electrical hazards such as thermal overload, over-current, and over-voltage. The following protective functions shall be provided:

- Both over and under voltage on DC and AC sides.
- Over-current on both DC and AC sides.
- Ensuring battery protection.
- Detection and management of internal faults such as over-temperature and logic failures.

Regarding scenario-1&2, the PCS shall comply with IEC 62477-1:

The power conversion system shall provide protection against various hazards as follows:

- The Power Conversion System (PCS) shall maintain protection against hazards under both normal operating conditions and single fault conditions, as outlined in this standard.
- The PCS must not pose any hazards during short circuit or overload scenarios. Sufficient protective systems or devices must be installed at appropriate locations to detect and either interrupt or limit the current flowing in any potential fault current path.
- The PCS must adhere to the protection requirements against electric shock as outlined in IEC 62477-1.
- The PCS shall provide both basic and fault protection and can be achieved by means of: Reinforced insulation, Protective separation between circuits, and any other Protection methods specified.
- The PCS shall provide Protection Against Fire and Thermal Hazards

The following table outlines various types of tests, whether they are type, routine sample tests, applicable to Power Conversion Systems (PCS) according to IEC 62477-1.

Test Type	Type	Routine	Sample
Visual inspection	X	X	
Mechanical tests			
Clearance and creepage distances test	X		
Non-accessibility test	X		

Test Type	Type	Routine	Sample
Ingress protection test (IP rating)	X		
Enclosure integrity test	X		
Deflection test	X		
Steady force test, 30N	X		
Steady force test, 250N	X		
Impact test	X		
Drop test	X		
Stress relief test	X		
Stability test	X		
Wall or ceiling mounted equipment test	X		
Handles and manual control securement test	X		
Electrical tests			
Impulse voltage test	X	X	X
a.c. or d.c. voltage test	X	X	
Partial discharge test	X	X	X
Protective impedance test	X	X	
Touch current measurement test	X		
Capacitor discharge test			
Limited power source test			
Temperature rise test			
Protective equipotential bonding test	X	X	
Abnormal operation tests			
Output Short circuit test	X		
Output overload test	X		
Breakdown of components test	X		
PWB short circuit test	X		
Loss of phase test	X		
Cooling failure tests	X		
Inoperative blower test	X		
Clogged filter test	X		
Loss of coolant test	X		

Test Type	Type	Routine	Sample
Material tests			
High current arcing ignition test	X		
Glow-wire test	X		
Hot wire ignition test	X		
Flammability test	X		
Flaming oil test	X		
Cemented joints test	X		
Environmental tests	X		
Dry heat test	X		
Damp heat test	X		
Vibration test	X		
Salt mist test	X		
Dust and sand test	X		
Hydrostatic pressure test	X	X	

Regarding scenario-3, the PCS shall comply with IEC 62109: Standard for Inverters, Converters, Controllers and Interconnection System Equipment for Use with Distributed Energy Resource

5.1.3 BATTERY MANAGEMENT SYSTEM (BMS):

The BMS is the brain of the battery system, with its primary function being to safeguard and protect the battery from damage in various operational scenarios. To achieve this, the BMS has to ensure that the battery operates within predetermined ranges for several critical parameters, including state of charge (SOC), state of health (SOH), voltage, temperature, and current. The BMS constantly monitors the status of the battery and uses application-specific algorithms to analyze the data, control the battery's environment, and balance it.

The battery management system shall incorporate the following monitoring, maintain optimal operation, and protect functions for Battery Cell/Rack/System:

- **Monitoring and Measurement:**

- Monitor individual battery cell voltages and overall module/tray voltage.
- Track battery cell temperatures at various points within the module/tray.
- Measure the current flowing through the battery module.

- **Cell Balancing:**
  - Ensure uniform voltage distribution among cells within the module/tray.
- **Protect battery cells and module/tray against:**
  - Over-charge protection to prevent excessive charging of the battery.
  - Over-discharge protection to prevent the battery from being discharged beyond safe levels.
  - Over-temperature protection mechanisms to prevent the battery from operating at temperatures that could damage it.
  - Over-current protection to safeguard against excessive current flow that could damage the battery.
  - Ground-fault detection to identify and mitigate any electrical faults that could occur.
  - Internal battery fault detection systems to identify and address any internal faults within the battery.
  - Overvoltage and under voltage conditions.
  - Short circuit occurrences.
  - Overheating or undercooling.

The BMS must align with the specified Safety Integrity Level (SIL) target. To assess charge control's impact on safety, battery system manufacturers should conduct the following tests according IEC 62619:

1- Overcharge Control of Voltage (Battery System)

Requirement: The BMS must regulate the charging voltage below the upper limit charging voltage of the cells.

2- Overcharge Control of Current (Battery System)

Requirement: If the input current to the cells and batteries exceeds the maximum charging current of the cells, the BMS shall interrupt the charging to protect the battery system from hazards associated with charging currents above the cells' specified maximum charging current.

3- Overheating Control (Battery System)

Requirement: The BMS should terminate charging when the temperature of the cells and/or battery exceeds the upper limit specified by the cell manufacturer. .

5.1.4 CONTROLLER:

Controller is the brain of the entire BESS. It monitors, controls, protects, communicates, and schedules the BESS's key components. As well as communicating with the components of the energy storage system itself, it can also communicate with external devices such as electricity meters and transformers, ensuring the (**BESS**) is operating optimally. The controller has multiple levels of protection, including overload protection in charging and reverse power protection in discharging. The controller can integrate with third-party SCADA and EMS for complete data acquisition and energy using the IEC 60870-5-104 Protocol, IEC 61850 ,Modbus, and IEC 60870-5-103 which operates on TCP/IP communication.

5.1.5 CONTAINER / KIOSK

- In case of mobile emergency battery energy system , container should be suitable for the BESS and the maintenance service with IP 54
- In case of mobile emergency battery energy system , it should be equipped with a manual tune frequency apparatus from (49 to 51 HZ) as option of …EDC
- In case of stationary battery energy storage system, kiosk should be with IP 54

5.1.6 HEATING, VENTILATION, AND AIR CONDITIONING (HVAC):

The HVAC is an integral part of a battery energy storage system; it regulates the internal environment by moving air between the inside and outside of the system's enclosure. With lithium battery systems maintaining an optimal operating temperature and good air distribution help prolong the cycle life of the battery system.

➢ Note: Liquid cooling system is acceptable also.

5.1.7 Fire Suppression:

The fire suppression system within a BESS is an additional layer of protection. If an elevated temperature outside the set parameters is reached, the BMS will automatically shut the system down; however, in the case of a thermal runaway, the BMS cannot be relied on as the only layer of protection. the fire suppression system will activate; this could be activated through gas, smoke, or heat detection, depending on which fire suppression system the BESS has. Once started, the fire suppression system will release an agent which suppresses the fire, providing a cooling effect and absorbing the heat.

The fire system may be Aerosol fire extinguishing system or Carbon dioxide fire extinguisher or Water firefighting system or combination of the above

5.1.8 Energy Management System (EMS):

The energy management system is in charge of controlling and scheduling BESS application activity. To schedule the various components on-site, the EMS communicates directly with the PCS/Hybrid Inverter and BMS, frequently considering external data points from things such as the electric grid, transformers, PV arrays, and loads. The EMS is responsible for determining when and how to discharge power, which is typically decided by the application specifics such as peak shaving, load shifting, or self-consumption. An EMS will optimize BESS performance by balancing application cycling data and battery life with the asset’s return on investment while at the same time considering the limitations of the BMS and PCS/Hybrid Inverter. The EMS will also collect and analyze BESS performance data, making reporting and forecasting easy.

6. VEHICLE SPECIFICATIONS (for mobile emergency battery energy storage vehicle):

- Maximum number of occupants: 2 persons.
- Top speed: 90 Km / h.
- Equipped with Hydraulic support system.
- Equipped with cable winding machine with adjustable motor
- Equipped with required cable of charging and discharging
- Firefighting system: Standard configuration.
- Ventilation and heat dissipation system: Standard configuration.
- Light system: Standard configuration.
- The vehicle commonly used in A.R.E. has an agent and a center for maintenance & spare parts within Greater Cairo and its suburbs.
- Submit all necessary documents for licensing.
- The roof, floor, and sides of container shall be electrical & thermal insulated.
- The vehicle should support the parking system.
- Container IP: 54
- The dimensions and weight of the vehicle shall be specified by …. EDC during tendering.

7. TECHNICAL SPECIFICATION OF SUPPLIED EQUIPMENT

Rated output power (AC)	186 KW / 372 KWH	250 KW / 500 KWH	500 KW / 1000 KWH
Rated input current (charging current)	282 A at 380 V	380 A at 380 V	760 A at 380 V
Rated output current (Load current)	282 A at 380 V	380 A at 380 V	760 A at 380 V
Charging cable cross section area (minimum) / insulation	4×(1×120mm2)CU or equal / double rubber (unarmoured)	4×(1×185mm2)CU or equal / double rubber (unarmoured)	4×(2×1×185mm2)CU or equa double rubber (unarmoured)
Load cable cross section area (minimum) / insulation	4×(1×120mm2)CU or equal / double rubber (unarmoured)	4×(1×185mm2)CU or equal / double rubber (unarmoured)	4×(2×1×185mm2)CU or equa double rubber (unarmoured)
Minimum length of each cable	30 m	30 m	30 m
Time of charging & discharging at rated output current	2h / 0.5 C	2h / 0.5 C	2h / 0.5 C
Life time of battery	≥ 5000 cycles	≥ 5000 cycles	≥ 5000 cycles
Efficiency after 5000 cycles	Decrease 10 %	Decrease 10 %	Decrease 10 %

Depth of discharge (DoD)	90 %	90 %	90 %
Load power factor range	0.8 Lag to 0.9 Lead		
AC system type	3 phases Low Voltage (VLL/Vph) 380/220 V - 4 wire system		
Option ratings according **EDC** requirements	600 KW for 1hr (discharging)	1000 KW for 1hr (discharging)	2000 KW for 1hr (discharging)

: **Spare parts (option):**

– The supplier shall supply spare parts for the maintenance of the BESS Vehicle to --EDC as needed for 10 years.

– Spare parts shall be described by the tenderer.

6. MAINTENANCE:

The supplier shall tender the criteria set for maintenance as follows:

1. ---EDC will do all necessary maintenance from the component replacement level all the way up to the module replacement level.
2. Maintenance aids such as diagnostic flow charts as well as adequate test functions shall be available.
3. The maintenance manual with circuit diagrams and part number to enable the above shall also be available in a well-written form.

7. INSPECTION AND TESTING:

Tests of all components shall be carried out by the manufacturer; a representative from the Distribution Company could attend these tests before acceptance.

8. TRAINING:

i. Four engineers shall inspect the BESS in the factory.

ii. Five engineers shall be trained on how to operate and maintain the BESS in the factory.

iii. All travel expenses including VISA, hotels, tickets, transportation, and hospitality are supported by the supplier.

9. CATALOGUES:

iv. All necessary drawings including layout, dimensions, single-line diagram of primary circuits, and the protection shall be submitted with delivery.

v. Technical catalogues for each equipment containing the technical data for all offered components of the unit shall be submitted.

vi. Technical catalogues for BESS.

vii. Maintenance manual for each equipment.

10. SUBMITALES:

viii. A certificate of origin country shall be provided for all equipment.

ix. Documents shall be supported with delivery for all equipment inside the BESS including the following data:

- General description.
- Theory of operation.
- Diagrams and circuits description.
- Detailed description and interconnection diagrams.

x. Result test in factory.

xi. Technical, operation, maintenance book, and services manuals precisely for all equipment.

xii. Operation guide.

xiii. Copy of software for the module system.

N.B.

- All documents should be in English.

11. GUARANTEE:

– The supplier shall guarantee the BESS with all equipment against all defects arising out of faulty design or workmanship, or of defective materials as follows: -

1. For the vehicle: 12 months from the delivery date or 100000 km which comes first.
2. For the BESS: Two years, the first one with a bid guarantee, and the second with a certificate.

– The good quality and manufacturing of the BESS shall be guaranteed.

– Defective or poorly assembled parts of the BESS Vehicle and its equipment shall be replaced or repaired promptly and without charge.